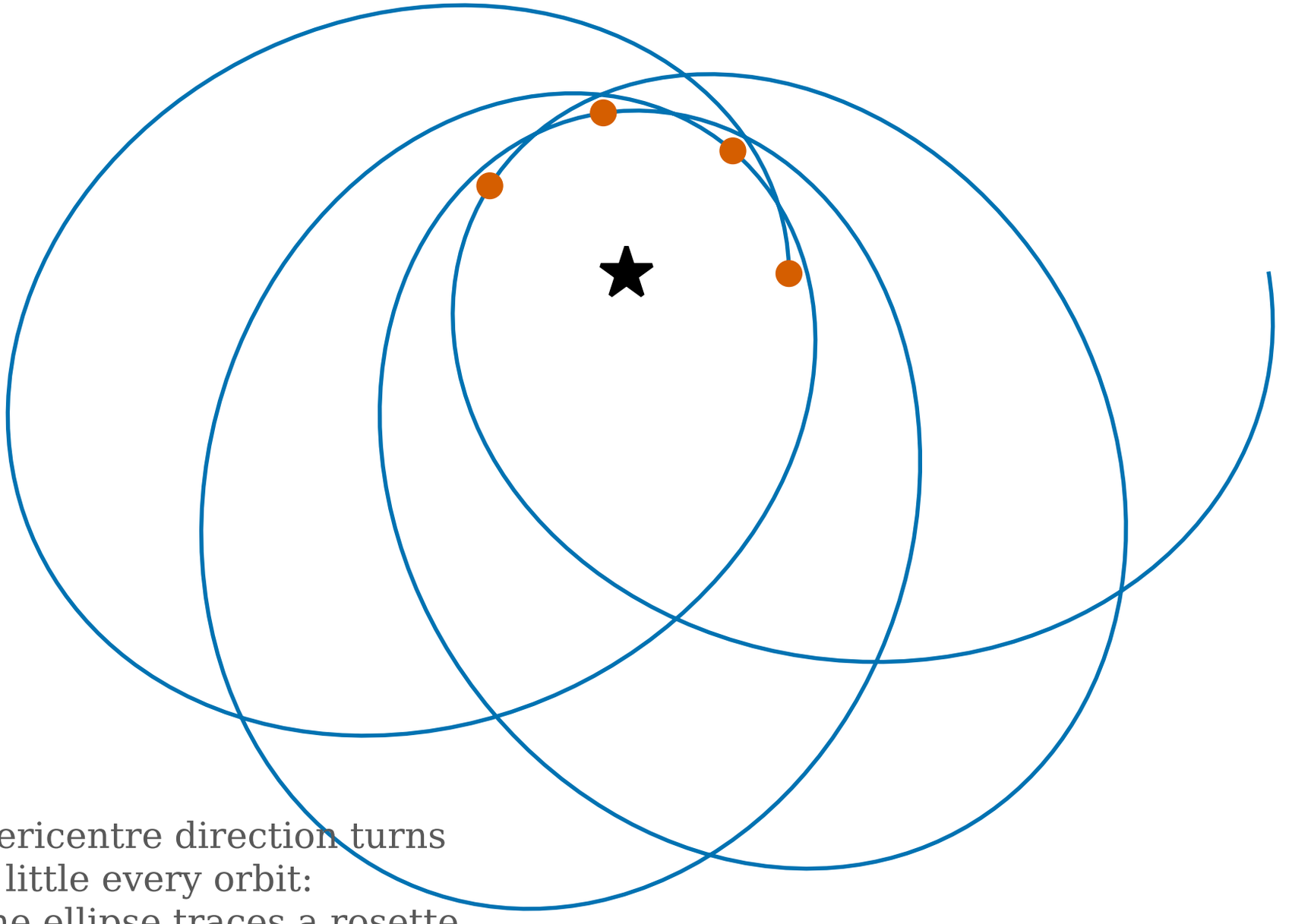

pericentre direction turns
a little every orbit:
the ellipse traces a rosette